ODZIV DISKRETNIH LINEARNIH SUSTAVA

→ zadani jednadžbama diferencija

* kod impulsnih odziva računamo da je sve prije bilo $\emptyset$

Impulsni odziv:

$u[n] = \delta[n]$

$y[n] = ?$

impulsni odziv: $h[n]$

$y[n] - 0.75y[n-1] = u[n]$ // jednadžba diferencija

$y[n] = +0.75\, y[n-1] + u[n]$ → * ima memoriju

→ s obzirom da je na ulazu samo 1 impuls, umjesto y pišemo $h[n]$, a umjesto $u[n] \to \delta[n]$

$h[n] = \delta[n] + 0.75\, h[n-1]$

$h[0] = \underbrace{\delta[0]}_{1} + 0.75\, \underbrace{h[-1]}_{\text{prije nule, ali rekli smo da je sve prije } 0 \Rightarrow 0} \Rightarrow h[0] = 1$

budući da smo rekli da su poč. uvjeti $\emptyset$, tj. da je sve prije nula 0

$h[1] = \underbrace{\cancel{\delta[1]}}_{0} + 0.75\, h[0] \Rightarrow h[1] = 0.75$

$h[2] = \underbrace{\cancel{\delta[2]}}_{0} + 0.75\, h[1] \Rightarrow h[2] = 0.75^2$

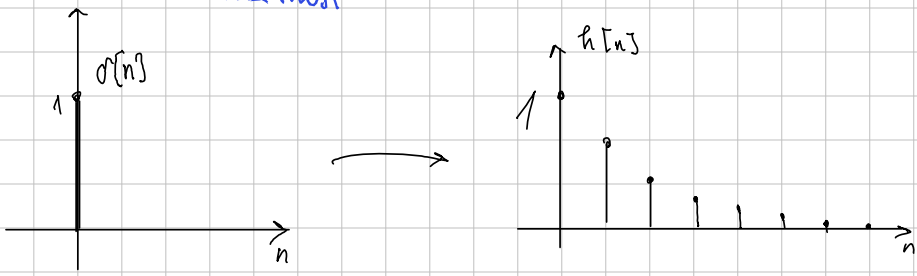
$h[3] = \delta[3] + 0.75 \cdot h[2] \Rightarrow h[3] = 0.75^3$

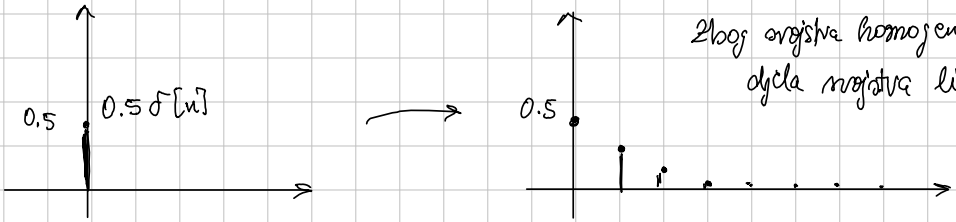
⋮

$h[n] = 0.75^n$ vrijedi za $n \geq 0$

→ pa još bolje $h[n] = 0.75^n \cdot u[n]$ impulsni odziv mirnog sustava

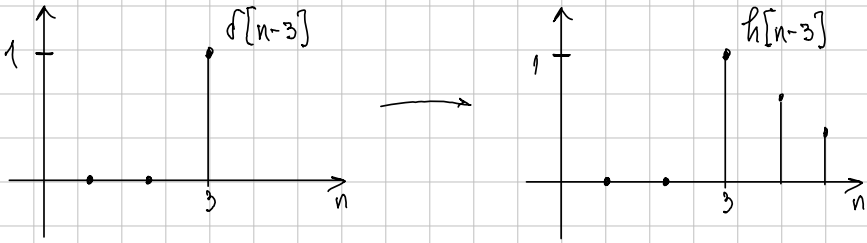
Linearnost





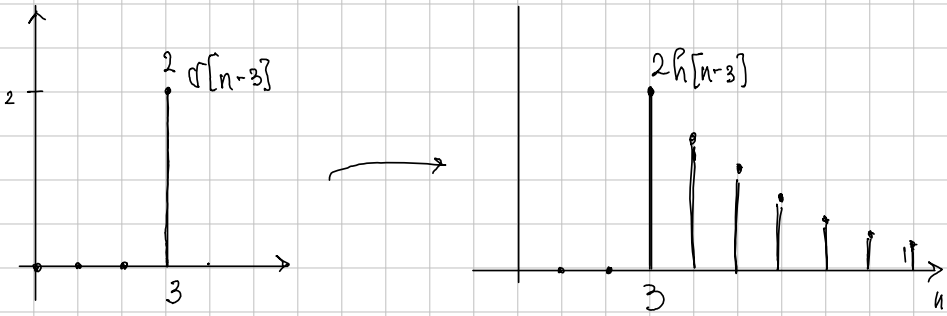
Zbog svojstva homogenosti kao dycla svojstva linearnosti!

Stalnost:

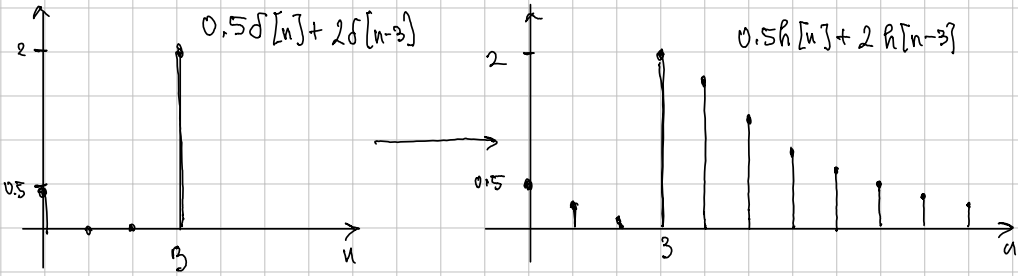


pomaknuti ulaz je dovoo do pomaknutog izlaza

Množenje s konstantom i pomak vrijediti istovremeno?



ili



video 49.

Impulsni odziv sa konvolucijom daje odziv MIRNOG SUSTAVA

Znači postoje 2 načina za pronalaženje

1. konvolucijom impulsnog odziva sa zadanom pobudom
2. Rješavanje homogeno-partikularne jed sa poč uvjetima da su $=\emptyset$

↳ treba davati jednako rješenje

Matlab 515.50 – 5:30

I. metoda → temeljena na DIREKTNOJ I. realizaciji

→ na ulazu jedan jedini impuls

→ poč. uvjeta nema; sve nula i mirno

→ jedan jedini impuls na ulazu

$$y[n] - 0.9\,y[n-1] + 0.2\,y[n-2] = u[n] + u[n-1] + u[n-2]$$

zanima nas impulsni odziv znači $u[n] = \delta[n]$

↳ $y[-1]=0$, $y[-2]=0$ i $y[n] \to h[n]$

$$h[n] - 0{,}9\,h[n-1] + 0.2\,h[n-2] = \delta[n] + \delta[n-1] + \delta[n-2]$$, za $n>2$ na ulazu nemamo ništa

$h[n] = C_1 \cdot 0{,}5^n + C_2 \cdot 0{,}4^n$; za $n>2$ ← → homogena

trebamo pronaći poč. uvjete:

moramo imati poč uvjete od „natrag prije"

$$h[n] = \delta[n] + \delta[n-1] + \delta[n-2] + 0{,}9\,h[n-1] - 0{,}2\,h[n-2]$$

$$h[0] = 1 + \emptyset + \emptyset + 0.9\,\underbrace{h[-1]}_{0} - 0.2\,\underbrace{h[-2]}_{0} \to \boxed{h[0] = 1}$$

$$h[1] = 0 + 1 + 0 + 0.9\,\underbrace{h[0]}_{1} - 0.2\,\underbrace{h[-1]}_{0} \to \boxed{h[1] = 1.9}$$

$$h[2] = \delta[2] + \delta[1] + \delta[0] + 0.9\,h[1] - 0.2\,h[0]$$

$$h[2] = 0 + 0 + 1 + 0.9 \cdot 1{,}9 - 0.2 \cdot 1 \to \boxed{h[2] = 2.51}$$

ovo su naši poč uvjeti!

→ u naše homogeno rješenje uvrštavamo $h[1]$ i $h[2]$

$$h[1] = C_1 \cdot 0{,}5^1 + C_2 \cdot 0{,}4^1 = 1.9$$

$$h[2] = C_1 \cdot 0{,}5^2 + C_2 \cdot 0{,}4^2 = 2.51$$

$$\boxed{C_1 = 35 \qquad C_2 = -39}$$

$$\rightsquigarrow h[n] = 35 \cdot 0{,}5^n - 39 \cdot 0{,}4^n, \quad n \geq 1$$

! ali želimo da naš impulsni odziv ide od $\emptyset$

$$\boxed{h[n] = (35 \cdot 0{,}5^n - 39 \cdot 0{,}4^n)\,u[n-1] + \delta[n]}$$

dodamo impuls „na silu"

II. metoda – na temelju direktne realizacije II

$$y[n]-0.9\,y[n-1]+0.2\,y[n-2]=u[n]+u[n-1]+u[n-2]$$

$$y[-1]=0 \qquad y[-2]=0 \qquad u[n]=\delta[n]$$

1. dio: ništa zakašnjelo, samo jedan ulaz: $y[n]-0{,}9\,y[n-1]+0.2\,y[n-2]=u[n]$

$$h[n]-0.9\cdot h[n-1]+0.2\,h[n-2]=\delta[n]$$

$n>0 \longrightarrow$ HOMOGENA: $h_a[n]-0.9\,h_a[n-1]+0.2\,h_a[n-2]=0$

$$h_a[n]=C_1\cdot 0{,}5^n+C_2\cdot 0{,}4^n$$

$$h_a[0]=\underbrace{0.9\,h[-1]}_{0}-\underbrace{0.2\,h[-2]}_{0}+\underbrace{\delta[0]}_{1} \longrightarrow h_a[0]=1$$

dovoljan poč. uvjet za računanje odziva jer je već uključen impuls

$h_a[-1]=C_1\cdot 0{,}5^{-1}+C_2\cdot 0{,}4^{-1}=0 \rightarrow$ jer je miran sustav

$h_a[0]=C_1+C_2=1 \rightarrow$ poč uvjet $\Rightarrow$ $\boxed{C_1[n]=5,\ C_2[n]=-4}$

$$h_a[n]=5\cdot 0{,}5^n-4\cdot 0{,}4^n \quad n\geq 0$$ (prvi pod sustav)

2. dio

$\rightarrow$ za dalje nas traži izlazna strana jednadžbe

$$h[n]=1\cdot \underbrace{h_a[n]}_{\text{umjesto } u[n]}+1\cdot \underbrace{h_a[n-1]}_{\text{umjesto } u[n-1]}+1\cdot h_a[n-2]$$

$$h[n]=5\cdot 0{,}5^n-4\cdot 0{,}4^n+5\cdot 0{,}5^{n-1}-4\cdot 0{,}4^{n-1}+5\cdot 0{,}5^{n-2}-4\cdot 0{,}4^{n-2}$$

$$h[n]=5\cdot 0{,}5^n(1+0{,}5^{-1}+0{,}5^{-2})-4\cdot 0{,}4^n(1+0{,}4^{n-1}+0{,}4^{n-2})$$

$$h[n]=35\cdot 0{,}5^n-39\cdot 0{,}4^n, \quad n\geq 2$$

nužno potrebno odrediti h(n) nekoliko koji fale

$$h[0]=h_a[0]+\underbrace{\cancel{h_a[-1]}}_{0}+\underbrace{\cancel{h_a[-2]}}_{0}=1$$

$$h_a[0]=5\cdot 0{,}5^0+4\cdot 0{,}4^0$$

$$h[1]=h_a[1]+\underbrace{h_a[0]}_{1}+\underbrace{\cancel{h_a[-1]}}_{0}=1.9$$

$$\boxed{h[n]=(35\cdot 0{,}5^n-39\cdot 0{,}4^n)\,u[n-1]+\delta[n]}$$

✓ isto je